Proof Outline

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1 Definitions and outline

Definition 1.1. *Suppose that $\Delta > 0$ and $L > 0$ are both independent of the problem dimension $d \geq 1$. Then, we denote $\mathcal{F}_d(\Delta, L)$ as the set of L-Lipschitz functions $g : \mathbb{R}^d \mapsto \mathbb{R}$ with the bounded function value $g(\mathbf{x}^0) - \inf_{\mathbf{x}\in\mathbb{R}^d} g(\mathbf{x}) \leq \Delta$.*

Definition 1.2. *Given a point $\mathbf{x} \in \mathbb{R}^d$ and $\delta > 0$, the δ-Goldstein subdifferential of a Lipschitz function g at $\mathbf{x}$ is given by*

$$\partial_\delta g(\mathbf{x}) := conv\left(\bigcup_{\mathbf{y}\in\mathbb{B}_\delta(\mathbf{x})} \partial g(\mathbf{y})\right).$$

The Goldstein subdifferential of f at $\mathbf{x}$ is the convex hull of the union of all generalized gradients at points in a δ-ball around $\mathbf{x}$. Accordingly, we can define the (δ, ϵ)-Goldstein stationary points; that is, a point $\mathbf{x} \in \mathbb{R}^d$ is a (δ, ϵ)-Goldstein stationary point if the following statement holds:

$$\min\{\|\mathbf{G}\| : \mathbf{G} \in \partial_\delta g(\mathbf{x})\} \leq \epsilon$$

Prove that the number of function value oracles required by our deterministic and stochastic gradient-free methods to find a (δ, ϵ)-Goldstein stationary point of $g \in \mathcal{F}_d(\Delta, L)$ is $O(\text{poly}(d, L, \Delta, 1/\epsilon, 1/\delta))$ when $\delta, \epsilon \in (0, 1)$ are constants

Proposition 1.3. *[LZJ22] Let $g_\delta(\mathbf{x}) = \mathbb{E}_{\mathbf{u}\sim\mathbb{P}}[g(\mathbf{x} + \delta\mathbf{u})]$ where $\mathbb{P}$ is an uniform distribution on a unit ball in ℓ_2-norm. Assuming that g is L-Lipschitz, we have (i) $|g_\delta(\mathbf{x}) - g(\mathbf{x})| \leq \delta L$, and (ii) g_δ is differentiable and L-Lipschitz with the $\frac{cL\sqrt{d}}{\delta}$-Lipschitz gradient where $c > 0$ is a constant. In addition, there exists a function g for which each of the above bounds are tight simultaneously.*

Lemma 1.4. *Let $G(x, \xi)$ be a stochastic process, where ξ represents randomness, that provides a noisy estimate of the gradient $\nabla g_\delta(x)$. Specifically, $G(x, \xi)$ satisfies the following properties. The estimator is unbiased for the gradient of the smoothed function, conditioned on x:*

$$\mathbb{E}[G(x, \xi) \mid x] = \nabla g_\delta(x)$$

The variance of the estimator, conditioned on x, is uniformly bounded:

$$\mathbb{E}\big[\|G(x, \xi) - \nabla g_\delta(x)\|^2 \mid x\big] \leq \sigma^2$$

for some constant $\sigma^2 \geq 0$.

Lemma 1.5 (Second-Moment Bound)**.** *Let $G(x, \xi)$ be an unbiased estimator of $\nabla g_\delta(x)$ with variance bounded by σ^2, i.e.*

$$\mathbb{E}\big[G(x, \xi) \mid x\big] = \nabla g_\delta(x), \quad \mathbb{E}\big[\|G(x, \xi) - \nabla g_\delta(x)\|^2 \mid x\big] \leq \sigma^2.$$

Then

$$\mathbb{E}\big[\|G(x, \xi)\|^2 \mid x\big] \;\leq\; \sigma^2 + \|\nabla g_\delta(x)\|^2.$$

Proof.

$$\begin{aligned}
\mathbb{E}\big[\|G(x,\xi)\|^2 \mid x\big] &= \mathbb{E}\big[\|G(x,\xi) - \nabla g_\delta(x) + \nabla g_\delta(x)\|^2 \mid x\big] && (1)\\
&= \mathbb{E}\big[\|G(x,\xi) - \nabla g_\delta(x)\|^2 \mid x\big] + \|\nabla g_\delta(x)\|^2 + 2\big\langle \mathbb{E}[G(x,\xi) - \nabla g_\delta(x) \mid x], \nabla g_\delta(x)\big\rangle && (2)\\
&= \mathbb{E}\big[\|G(x,\xi) - \nabla g_\delta(x)\|^2 \mid x\big] + \|\nabla g_\delta(x)\|^2 && (3)\\
&\le \sigma^2 + \|\nabla g_\delta(x)\|^2, && (4)
\end{aligned}$$

where we used the zero-mean property of the estimator to drop the cross-term and the variance bound in the last step. □

1.1 Outline

The proof proceeds in several steps:

1. **Establish a Descent Inequality:** Use the smoothness of f_δ and properties of $\mathbf{g}^t$ to relate the gradient norm to function value decreases.
2. **Sum Over Iterations:** Aggregate the inequality over all iterations to bound the average expected gradient norm.
3. **Bound the Function Value Gap:** Use properties of f_δ and the function class $\mathcal{F}_d(\Delta, L)$ to control the total decrease.
4. **Optimize the Stepsize:** Choose η to balance terms and simplify the bound.
5. **Relate to Goldstein Subdifferential:** Use Theorem 3.1 to connect ∇f_δ to $\partial_\delta f$ [Gol09].
6. **Compute Complexity:** Determine T to achieve the desired error and calculate the number of function evaluations.

2 Proof

2.1 Step 1 : Descent Inequality (Lemma D.2 [LZJ22])

Since g_δ is differentiable with a Lipschitz gradient (constant $\frac{cL\sqrt{d}}{\delta}$ per Proposition 1.1), for any step $x^{t+1} = x^t - \eta G^t$, we can use the descent property of gradient methods:

$$g_\delta(x^{t+1}) \le g_\delta(x^t) + \langle \nabla g_\delta(x^t), x^{t+1} - x^t\rangle + \frac{cL\sqrt{d}}{2\delta}\|x^{t+1} - x^t\|^2.$$

Substitute $x^{t+1} - x^t = -\eta G^t$:

$$g_\delta(x^{t+1}) \le g_\delta(x^t) - \eta\langle \nabla g_\delta(x^t), G^t\rangle + \frac{cL\sqrt{d}}{2\delta}\eta^2\|G^t\|^2.$$

Take the conditional expectation given x^t:

$$\mathbb{E}[g_\delta(x^{t+1})|x^t] \le g_\delta(x^t) - \eta\langle \nabla g_\delta(x^t), \mathbb{E}[G^t|x^t]\rangle + \frac{cL\sqrt{d}}{2\delta}\eta^2\mathbb{E}[\|G^t\|^2|x^t].$$

Since $\mathbb{E}[G^t|x^t] = \nabla g_\delta(x^t)$ (Proposition 1.2), and Lemma 1.1 also provides $\mathbb{E}[\|G^t\|^2|x^t] \le \sigma^2 + \|\nabla g_\delta(x)\|^2$, we get:

$$\mathbb{E}[g_\delta(x^{t+1})|x^t] \le g_\delta(x^t) - \eta\|\nabla g_\delta(x^t)\|^2 + \eta^2\frac{cL\sqrt{d}}{2\delta}\cdot\left(\sigma^2 + \|\nabla g_\delta(x)\|^2\right).$$

Take full expectation and rearrange:

$$\mathbb{E}[\|\nabla g_\delta(x^t)\|^2] \le \frac{1}{1 - \frac{\eta c\sqrt{d}L}{2\delta}}\left(\frac{\mathbb{E}[g_\delta(x^t)] - \mathbb{E}[g_\delta(x^{t+1})]}{\eta} + \frac{\eta c\sqrt{d}L}{2\delta}\sigma^2\right).$$

This is the key descent inequality from Lemma D.2.

2.2 Step 2: Sum Over Iterations

Sum this inequality from $t = 0$ to $T - 1$:

$$\sum_{t=0}^{T-1} \mathbb{E}[\|\nabla g_\delta(x^t)\|^2] \leq \sum_{t=0}^{T-1} \left(\frac{1}{1 - \frac{\eta c\sqrt{d}L}{2\delta}} \left(\frac{\mathbb{E}[g_\delta(x^t)] - \mathbb{E}[g_\delta(x^{t+1})]}{\eta} + \frac{\eta c\sqrt{d}L}{2\delta}\sigma^2 \right) \right).$$

The left-hand side is the total expected squared gradient norm. For the right-hand side:

- The function value terms telescope: $\sum_{t=0}^{T-1}(\mathbb{E}[g_\delta(x^t)] - \mathbb{E}[g_\delta(x^{t+1})]) = g_\delta(x^0) - \mathbb{E}[g_\delta(x^T)]$.
- The second term sums over T iterations: $\sum_{t=0}^{T-1} \frac{\eta c\sqrt{d}L\sigma^2}{2\delta} = \frac{\eta c\sqrt{d}L\sigma^2 T}{2\delta}$.

So:

$$\sum_{t=0}^{T-1} \mathbb{E}[\|\nabla g_\delta(x^t)\|^2] \leq \frac{1}{1 - \frac{\eta c\sqrt{d}L}{2\delta}} \left(\frac{g_\delta(x^0) - \mathbb{E}[g_\delta(x^T)]}{\eta} + \frac{\eta c\sqrt{d}L}{2\delta}\sigma^2 \right)$$

2.3 Step 3: Bound the Function Value Gap

We need to bound $g_\delta(\mathbf{x}^0) - \mathbb{E}[g_\delta(\mathbf{x}^T)]$:

- **Upper Bound:** By Proposition 1.1, $g_\delta(\mathbf{x}^0) \leq g(\mathbf{x}^0) + \delta L$.
- **Lower Bound:** Since $g_\delta(\mathbf{x}) = \mathbb{E}_{\mathbf{u}}[g(\mathbf{x} + \delta\mathbf{u})]$ and g is bounded below, $g_\delta(\mathbf{x}) \geq \inf_{\mathbf{y}} g(\mathbf{y})$ for all $\mathbf{x}$. Thus, $\mathbb{E}[g_\delta(\mathbf{x}^T)] \geq \inf_{\mathbf{x}} g(\mathbf{x})$.
- **Function Class:** Since $g \in \mathcal{F}_d(\Delta, L)$, we have $g(\mathbf{x}^0) - \inf_{\mathbf{x}} g(\mathbf{x}) \leq \Delta$.

Combine these:

$$g_\delta(\mathbf{x}^0) - \mathbb{E}[g_\delta(\mathbf{x}^T)] \leq g(\mathbf{x}^0) + \delta L - \inf_{\mathbf{x}} g(\mathbf{x}) \leq \Delta + \delta L.$$

Substitute into and divide by T:

$$\frac{1}{T}\sum_{t=0}^{T-1} \mathbb{E}[\|\nabla g_\delta(\mathbf{x}^t)\|^2] \leq \frac{\Delta + \delta L}{\eta T\left(1 - \frac{\eta c\sqrt{d}L}{2\delta}\right)} + \eta \cdot \frac{\frac{\eta c\sqrt{d}L\sigma^2}{2\delta}}{1 - \frac{\eta c\sqrt{d}L}{2\delta}}.$$

2.4 Step 4: Optimize the Stepsize

To the RHS of the above summation as small as possible so that it exactly balances descent term against variance term yielding optimal rate.

$$\frac{\Delta + \delta L}{\eta T\left(1 - \frac{\eta c\sqrt{d}L}{2\delta}\right)} = \eta \cdot \frac{\frac{\eta c\sqrt{d}L\sigma^2}{2\delta}}{1 - \frac{\eta c\sqrt{d}L}{2\delta}}.$$

$$\eta = \sqrt{\frac{2\delta\,(\Delta + \delta)}{T\sigma^2 c\sqrt{d}L}}$$

Now using this we rewrite the summation term and simplifying

$$\frac{1}{T}\sum_{t=0}^{T-1} \mathbb{E}[\|\nabla g_\delta(\mathbf{x}^t)\|^2] \leq \frac{2\,(\Delta + \delta)}{T\eta\left(\sqrt{\frac{2\delta(\Delta+\delta)}{T\sigma^2 c\sqrt{d}L}} - \frac{\Delta+\delta}{T\sigma^2}\right)}$$

Since the random count R $\in$ {0, 1, 2, . . . , T-1} is uniformly sampled, we have

$$\mathbb{E}[\|\nabla g_\delta(\mathbf{x}^R)\|^2] = \frac{1}{T}\sum_{t=0}^{T-1} \mathbb{E}[\|\nabla g_\delta(\mathbf{x}^t)\|^2] \leq \frac{2\,(\Delta + \delta)}{T\eta\left(\sqrt{\frac{2\delta(\Delta+\delta)}{T\sigma^2 c\sqrt{d}L}} - \frac{\Delta+\delta}{T\sigma^2}\right)}$$

2.5 Step 5: Relate to Goldstein Subdifferential

By Theorem 3.1 [LZJ22], $\nabla g_\delta(\mathbf{x}^R) \in \partial_\delta g(\mathbf{x}^R)$, so:

$$\min\{\|G\| : G \in \partial_\delta g(\mathbf{x}^R)\} \leq \|\nabla g_\delta(\mathbf{x}^R)\|.$$

Take expectations:

$$\mathbb{E}[\min\{\|G\| : G \in \partial_\delta g(\mathbf{x}^R)\}] \leq \mathbb{E}[\|\nabla g_\delta(\mathbf{x}^R)\|].$$

By Jensen's inequality, $\mathbb{E}[\|\nabla g_\delta(\mathbf{x}^R)\|] \leq \sqrt{\mathbb{E}[\|\nabla g_\delta(\mathbf{x}^R)\|^2]}$, so:

$$\mathbb{E}[\|\nabla g_\delta(\mathbf{x}^R)\|] \leq \sqrt{\frac{2\left(\Delta+\delta\right)}{T\eta\left(\sqrt{\frac{2\delta(\Delta+\delta)}{T\sigma^2 c\sqrt{d}L}} - \frac{\Delta+\delta}{T\sigma^2}\right)}}$$

$$\mathbb{E}[\min\{\|G\| : G \in \partial_\delta g(\mathbf{x}^R)\}] \leq \sqrt{\frac{2\left(\Delta+\delta\right)}{T\eta\left(\sqrt{\frac{2\delta(\Delta+\delta)}{T\sigma^2 c\sqrt{d}L}} - \frac{\Delta+\delta}{T\sigma^2}\right)}}$$

2.6 Setp 5: Compute Complexity

$$\sqrt{\frac{2\left(\Delta+\delta\right)}{T\eta\left(\sqrt{\frac{2\delta(\Delta+\delta)}{T\sigma^2 c\sqrt{d}L}} - \frac{\Delta+\delta}{T\sigma^2}\right)}} \leq \epsilon$$

$$T \geq \frac{\left(\Delta+\delta L\right)c\sqrt{d}L}{2\delta\sigma^2} \cdot \left(\frac{2\sigma^2}{\epsilon^2}+1\right)^2$$

Now each gradOL epoch uses $\mathcal{O}(K)$ compleixty for inner minimization. So overall complexity becomes

$$\boxed{T = \mathcal{O}\Big(\frac{\left(\Delta+\delta L\right)c\sqrt{d}L}{2\delta\sigma^2} \cdot \left(\frac{2\sigma^2}{\epsilon^2}+1\right)^2 K\Big)}$$

References

[Gol09] Herbert Goldstein. "Subdifferential calculus and monotonicity". In: *Mathematics of Operations Research* 34.3 (2009), pp. 451–469.

[LZJ22] Tianyi Lin, Zeyu Zheng, and Michael I. Jordan. "Gradient-Free Methods for Deterministic and Stochastic Nonsmooth Nonconvex Optimization". In: *Advances in Neural Information Processing Systems*. Vol. 35. 2022. URL: `https://proceedings.neurips.cc/paper_files/paper/2022/hash/a78f142aec481e68c75276756e0a0d91-Abstract-Conference.html`.